

2024



Driving Patient-Centric Innovation in Life Sciences Using Generative AI with Pfizer

Learn how Pfizer and AWS are collaborating to build generative AI prototypes to help bring innovative therapeutics to market quickly.

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Cut time

from prototype to MVP from 3+ months to 6 weeks

16,000 hours

of search time saved annually

55% reduction

in infrastructure costs



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Overview

With a mission to transform healthcare for its 1.3 billion patients, [Pfizer](#) is always seeking to i edge technology. To optimize the development of new medicines through cloud-based solutio Amazon Web Services (AWS) have created the Pfizer-Amazon Collaboration Team (PACT) initiative.



Under PACT, Pfizer has pursued 14 projects, including generative artificial intelligence (AI) and machine learning (ML) to save scientists up to 16,000 hours of search time annually and cut infrastructure costs by 55 percent. Pfizer uses a plethora of tools from AWS to explore new ideas, quickly build prototypes, and drive a passion for innovation in digital drug development.



Opportunity | Supporting Rapid Innovation through the PACT Initiative

Pfizer has focused on scientific innovation as a core value since it was founded in 1849. “To keep up with the pace of technology, somebody would constantly have to learn the technologies that AWS releases,” says Vijay Bulusu, Pfizer’s head of data and digital innovation for Pharmaceutical Sciences Small Molecule (PSSM). “Our collaboration with AWS lets Pfizer remain focused on the science yet use the breadth and depth of new technologies that AWS brings to the table

In 2021, Pfizer’s PSSM group and AWS launched PACT, applying AWS capabilities in analytics, ML, compute, storage, security, and cloud data warehousing to Pfizer laboratory, clinical manufacturing, and clinical supply chain efforts. Pfizer teams in pharmaceutical development and clinical manufacturing hatch promising ideas, while AWS provides technical expertise through solutions architects and the AWS Prototyping and Cloud Engineering team.

With Pfizer teams highly focused on ongoing technology projects, PACT addresses challenges around the bandwidth needed for prototyping. “With this collaboration, AWS brings in its prototyping teams to work with business and IT teams directly on relevant challenges,” says Bulusu. “This gave us the ability to prototype rapidly, with a learn, inform, and fail fast cycle.”

Successful prototypes move to the minimum viable product (MVP) stage and possibly on to production, under the guidance of [AWS Professional Services](#), a global team of experts that can help organizations realize desired business outcomes on AWS. “The great value for us has been not having to use internal bandwidth for evaluation and prototyping, which requires a lot of deep technical knowledge,” Bulusu says. “Working alongside AWS, we can test new technologies to solve challenges and scale a solution without having to become experts on a wide variety of technologies that we may or may not end up using.”

Solution | Saving Scientists up to 16,000 Hours of Search Time per Year Using ML and Generative AI

With PACT, Pfizer has cut significant time for a prototype to get to MVP. “If we were to do this ourselves, assuming we could find the bandwidth and people with appropriate skills, it would take at least 3 months,” says Bulusu. “A PACT prototype typically has been no more than 6 weeks.” PACT selects use cases based on the expected business value of the innovation, focusing on problems that span internal businesses.

The first use case sought to minimize the time spent in data discovery upto 80 percent for 1,500 PSSM scientists. The development of one drug can result in approximately 20,000 documents, and scientists often must look for data manually using a variety of tools to find historical data. Early experimentation led to the development of a new AWS service: [Amazon Kendra](#), an intelligent enterprise search service that organizations use to search across different content repositories with built-in connectors.

For greater efficiency in search capabilities, PACT teams now use generative AI, accessing Anthropic’s Claude 2.1 through Amazon Bedrock, a fully managed service that offers a choice of high-performing foundation models from leading AI companies. Scientists access [Amazon Bedrock](#) through an internal platform called Vox to search documents by voice command and chatbot, asking questions in natural language. Pfizer estimates that, annually, scientists could save up to 16,000 hours of searching and extracting data. It also estimates a 55 percent reduction in infrastructure costs.

Another PACT prototype detects anomalies in Pfizer’s Portable Continuous Miniature and Modular (PCMM) manufacturing, a continuous manufacturing process for producing solid oral-dose medicines. Engineers built the PCMM anomaly detection prototype using [Amazon SageMaker](#), a fully managed service that brings together a broad set of tools for high-performance, low-cost ML for virtually any use case.

Pfizer scientists can detect anomalies in PCMM manufacturing as they occur, predict maintenance needs, and reduce equipment downtime through the solution’s ML capabilities built on AWS. The solution also uses [Amazon Lookout for Equipment](#), an ML industrial equipment monitoring service that detects abnormal equipment behavior so that organizations can act and avoid unplanned downtime, and [Amazon Lookout for Metrics](#), which automatically detects anomalies within metrics and identifies their root causes. Eventually, this work led to the implementation of a manufacturing analytics tool that specializes in time series data. “Our large molecule group now uses the technology too,” says Bulusu. “Learnings from PACT projects have sparked innovation in multiple Pfizer groups and business lines.”

PACT encourages the passion of Pfizer teams even if their ideas are not production ready. For example, the group explored the idea of using virtual and augmented reality for training. “That’s a great example of doing a fast prototype and understanding that users were not completely prepared for this technology yet,” Bulusu says. “We can learn fast and fail fast with these prototyping experiments, and that’s a major benefit for us.”

Outcome | Nurturing a Company-Wide Culture of In



From PACT's initial list of 14 projects, five ideas have moved into production so far. "Even though we are just doing prototyping and there's no immediate revenue in it for AWS, AWS always brings forward its top teams," says Bulusu. "We've always been impressed with the talent that we work with from AWS."

The PACT initiative has emboldened the PSSM group to propose ideas that it previously might have considered unfeasible. "With access to the talents and technologies of AWS, we've changed our innovation culture and done a lot in a very short time," says Bulusu. "As soon as people started seeing the case studies that we shared across groups and hearing about our collaboration with the AWS team, we started hearing about challenges that people wouldn't raise their hands for previously. The most important benefit I've seen through the PACT program is that it has inspired people to think bolder."

About Pfizer

An American multinational pharmaceutical and biotechnology corporation, Pfizer applies science and its global resources to provide people with therapies that extend and significantly improve their lives.

AWS Services Used

Amazon Kendra

Amazon Kendra offers an intelligent enterprise search solution that increases employee productivity and improves customer satisfaction.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

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Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies through a single API, along with a broad set of capabilities to build generative AI applications with security, privacy, and responsible AI.



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